CHAPTER 18

DYNAMICAL SYSTEMS ON NETWORKS

A discussion of dynamical systems on networks, a subject area that is in its infancy but about which we nonetheless have some interesting results

THE epidemic models of Chapter 17 are a particular example of the more general concept of dynamical systems on networks. A dynamical system is any system whose state, as represented by some set of quantitative variables, changes over time according to some given rules or equations. Dynamical systems come in continuous- and discrete-time varieties and can be either deterministic or stochastic. The epidemic models we looked at, for instance, were continuous-time dynamical systems because their equations described the continuous-time variation of the variables. They were also deterministic because the equations we wrote down exactly determine the values of all variables for all time: there was no random or external element affecting the evolution whose value was not known in advance. On the other hand, an explicit computer simulation of, say, an SI epidemic model on a network would be a stochastic dynamical system and might use either continuous- or discrete-time. The stochastic element in this case corresponds to the chance infection of a susceptible individual by an infectious neighbor. And time might be represented in discrete time-steps, although it might not, depending on the decision of the researcher.

Many other real-world processes—or simplified models of real-world processes—can be represented as dynamical systems on networks. The spread of news or information between friends, the movement of money through an economy, the flow of traffic on roads, data over the Internet, or electricity over the grid, the evolution of populations in an ecosystem, the changing concentrations of metabolites in a cell, and many other systems of scientific interest are best thought of as dynamical processes of one kind or another taking place on an appropriate network.

In other, non-network contexts, the theory of dynamical systems is a well-developed branch of mathematics and physics. (See, for example, the book by Strogatz [307].) In this chapter we delve into some of this theory and show how it can be applied to dynamical systems on networks. Necessarily our introduction only skims the surface of what could be said; dynamical systems is a topic of entire books in its own right. But the material covered here gives a flavor of the kinds of calculation that are possible.

18.1 DYNAMICAL SYSTEMS

Our discussion in this chapter will concentrate principally on deterministic systems of continuous real-valued variables evolving in continuous time t. We begin by introducing some of the basic ideas in a non-network context, then we extend these ideas to networks.

A simple (non-network) example of a continuous dynamical system is a system described by a single real variable $x(t)$ that evolves according to a first-order differential equation

$$\frac{dx}{dt} = f(x), \tag{18.1}$$

where $f(x)$ is some specified function of x. Typically we will also give an initial condition that specifies the value x_0 taken by x at some initial time t_0.

The fully-mixed SI model of Section 17.2 is an example of a dynamical system of this kind, having a single variable x representing the fraction of infected individuals in the system, obeying the equation

$$\frac{dx}{dt} = \beta x(1-x). \tag{18.2}$$

(See Eq. (17.5).) Thus in this case we have $f(x) = \beta x(1-x)$.

One can also have dynamical systems of two variables:

$$\frac{dx}{dt} = f(x,y), \qquad \frac{dy}{dt} = g(x,y), \tag{18.3}$$

and the approach can be extended to larger numbers of variables as well. When we come to consider systems on networks we will put separate variables on each vertex of the network.

One could also imagine making the functions on the right-hand sides of our equations depend explicitly on time t:

$$\frac{dx}{dt} = f(x,t).$$

(18.4)

This, however, can be regarded as merely a special case of Eq. (18.3). If we write

$$\frac{dx}{dt} = f(x,y), \qquad \frac{dy}{dt} = 1,$$

(18.5)

with initial condition $y(0) = 0$, then we have $y = t$ for all times and $dx/dt = f(x, t)$ as required. By this trick it is always possible to turn equations with explicit dependence on t into equations without explicit dependence on t but with one extra variable. For this reason we will confine ourselves in this chapter to systems with no explicit dependence on t.

Another possible generalization would be to consider systems governed by equations containing higher derivatives, such as second derivatives. But these can also be reduced to simpler cases by introducing extra variables. For instance the equation

$$\frac{d^2x}{dt^2} + \left(\frac{dx}{dt}\right)^2 - \frac{dx}{dt} = f(x),$$

(18.6)

can be transformed by introducing a new variable $y = dx/dt$ so that we have

$$\frac{dx}{dt} = y, \qquad \frac{dy}{dt} = f(x) - y^2 + y,$$

(18.7)

which is a special case of Eq. (18.3) again.

Thus the study of systems of equations like (18.1) and (18.3) covers a broad range of situations of scientific interest. Let us look at some of the techniques used to analyze such equations.

18.1.1 FIXED POINTS AND LINEARIZATION

Equation (18.1), which involves only the one variable x, can, at least in principle, always be solved by simply rearranging and integrating:

$$\int_{x_0}^{x} \frac{dx'}{f(x')} = t - t_0,$$

(18.8)

although in practice the integral may not be known in closed form. For cases with two or more variables, on the other hand, it is not in general possible to find a solution. And for the network examples that we will be studying shortly the number of variables is typically very large, so that, unless we are lucky (as we were with some of the epidemiological models of the previous chapter), full analytic solutions are unlikely to be forthcoming.

We can of course integrate the equations numerically and in some cases this can give useful insight. But let's not give up on analytic approaches yet. There is in fact a well-developed set of techniques for understanding how dynamical systems work without first solving their equations exactly. Most of those techniques focus on the properties of fixed points.

A *fixed point* is a steady state of the system—any value of the variable or variables for which the system is stationary and doesn't change over time. In the one-variable system, Eq. (18.1), for example, a fixed point is any point $x = x^*$ for which the function on the right-hand side of the equation is zero

$$f(x^*) = 0,$$

(18.9)

so that $dx/dt = 0$ and x doesn't move. If, in the evolution of the system, x ever reaches a fixed point then it will remain there forever. The fixed points of a one-variable system can be found simply by solving Eq. (18.9) for x.

In a two-variable system like Eq. (18.3) a fixed point is a pair of values (x^*, y^*) such that $f(x^*, y^*) = 0$ *and* $g(x^*, y^*) = 0$, making $dx/dt = dy/dt = 0$ so that both variables stand still at this point.

Consider the SI model of Eq. (18.2). Putting $f(x) = 0$ in this model gives us $\beta x(1 - x) = 0$, which has solutions $x = 1$ and $x = 0$ for the fixed points. We can see immediately what these fixed points mean in epidemiological terms. The first at $x = 1$ represents the steady state in which everyone in the system is infected. Clearly once everyone is infected the system doesn't change any more, because there is no one else to infect and because in the SI model no one recovers either. The second fixed point $x = 0$ corresponds to the state of the system where no one is infected. In this state no one will ever become infected, since there is no one to catch the disease from, so again we have a steady state.

The importance of fixed points in the study of dynamical systems derives from two key features of these points: first, they are relatively easy to find, and second, it is straightforward to determine

the dynamics of the system when it is close to, but not exactly at, a fixed point. The dynamics close to a fixed point is found by expanding about the point as follows.

Consider first a simple one-variable system obeying Eq. (18.1). We represent the value of x close to a fixed point at x^* by writing $x = x^* + \epsilon$ where ϵ, which represents our distance from the fixed point, is small. Then

$$\frac{dx}{dt} = \frac{d\epsilon}{dt} = f(x^* + \epsilon).$$

(18.10)

Now we perform a Taylor expansion of the right-hand side about the point $x = x^*$ to get

$$\frac{d\epsilon}{dt} = f(x^*) + \epsilon f'(x^*) + O(\epsilon^2),$$

(18.11)

where f' represents the derivative of f with respect to its argument. Neglecting terms of order ϵ^2 and smaller and noting that $f(x^*) = 0$ (see Eq. (18.9)), we then have

$$\frac{d\epsilon}{dt} = \epsilon f'(x^*).$$

(18.12)

This is a linear first-order differential equation with solution

$$\epsilon(t) = \epsilon(0)\, e^{\lambda t},$$

(18.13)

where

$$\lambda = f'(x^*).$$

(18.14)

Note that λ is just a simple number, which we can calculate provided we know the position x^* of the fixed point and the function $f(x)$. Depending on the sign of λ, Eq. (18.13) tells us that our distance $\in$ from the fixed point will either grow or decay exponentially in time. Thus this analysis allows us to classify our fixed points into two types. An *attracting fixed point* is one with $\lambda < 0$, for which points close by are attracted towards the fixed point and eventually flow into it. A *repelling fixed point* is one with $\lambda > 0$, for which points close by are repelled away. In between these two types there is a special case when $\lambda = 0$ exactly. Fixed points with $\lambda = 0$ are usually still either attracting or repelling,[267] but one cannot tell which is which from the analysis here; one must retain some of the higher-order terms that we dropped in Eq. (18.11) to determine what happens.

Analysis of the kind represented by Eq. (18.12) is known as *linear stability analysis*. It can be applied to systems with two or more variables as well. Consider, for instance, a dynamical system governed by equations of the form of Eq. (18.3), with a fixed point at (x^*, y^*), meaning that

$$f(x^*, y^*) = 0, \qquad g(x^*, y^*) = 0.$$

(18.15)

We represent a point close to the fixed point in the two-dimensional x, y space by $x = x^* + \in_x$ and $y = y^* + \in_y$, where $\in_x$ and $\in_y$ are both assumed small. As before we expand about the fixed point, performing now a double Taylor expansion:

$$\begin{aligned}\frac{dx}{dt} = \frac{d\epsilon_x}{dt} &= f(x^* + \epsilon_x, y^* + \epsilon_y) \\ &= f(x^*, y^*) + \epsilon_x f^{(x)}(x^*, y^*) + \epsilon_y f^{(y)}(x^*, y^*) + \ldots,\end{aligned}$$

(18.16)

where $f^{(x)}$ and $f^{(y)}$ indicate the derivatives of f with respect to x and y. Making use of Eq. (18.15) and neglecting all higher-order terms in the expansion, we can simplify this expression to

$$\frac{d\epsilon_x}{dt} = \epsilon_x f^{(x)}(x^*, y^*) + \epsilon_y f^{(y)}(x^*, y^*).$$

(18.17)

Similarly

$$\frac{d\epsilon_y}{dt} = \epsilon_x g^{(x)}(x^*, y^*) + \epsilon_y g^{(y)}(x^*, y^*).$$

(18.18)

We can combine Eqs. (18.17) and (18.18) and write them in matrix form as

$$\frac{d\epsilon}{dt} = \mathbf{J}\epsilon,$$

(18.19)

where $\in$ is the two-component vector ($\in_x$, $\in_y$) and **J** is the Jacobian matrix

$$\mathbf{J} = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{pmatrix},$$

(18.20)

where the derivatives are all evaluated at the fixed point.

For systems of three or more variables we an employ the same approach and again arrive at Eq. (18.19), but with the rank of the vectors and matrices increasing with increasing number of variables.

Equation (18.19) is again a linear first-order differential equation but its solution is more complicated than for the one-variable equivalent. Let us begin with a particular simple case, the case where the Jacobian matrix is diagonal:

$$\begin{pmatrix} \frac{d\epsilon_x}{dt} \\ \frac{d\epsilon_y}{dt} \end{pmatrix} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \begin{pmatrix} \epsilon_x \\ \epsilon_y \end{pmatrix},$$

(18.21)

where λ_1 and λ_2 are real numbers. In this case, the equations for $\in_x$ and $\in_y$ separate from one another thus:

$$\frac{d\epsilon_x}{dt} = \lambda_1\epsilon_x, \qquad \frac{d\epsilon_y}{dt} = \lambda_2\epsilon_y,$$

(18.22)

and we can solve them separately to get

$$\epsilon_x(t) = \epsilon_x(0)\,e^{\lambda_1 t}, \qquad \epsilon_y(t) = \epsilon_y(0)\,e^{\lambda_2 t},$$

(18.23)

or equivalently

$$x(t) = x^* + \epsilon_x(0)\,e^{\lambda_1 t}, \qquad y(t) = y^* + \epsilon_y(0)\,e^{\lambda_2 t},$$

(18.24)

so that x and y are independently either attracted or repelled from the fixed point over time, depending on the signs of the two quantities

$$\lambda_1 = \left(\frac{\partial f}{\partial x}\right)_{\substack{x=x^*\\y=y^*}}, \qquad \lambda_2 = \left(\frac{\partial g}{\partial y}\right)_{\substack{x=x^*\\y=y^*}}.$$

(18.25)

These results give rise to a variety of possible behaviors of the system near the fixed point, as shown in Fig. 18.1. If λ_1 and λ_2 are both negative, for instance, then the fixed point will be attracting, while if they are both positive it will be repelling. If they are of opposite signs then we have a new type of point called a *saddle point* that attracts along one axis and repels along the other. In some respects a saddle point is perhaps best thought of as a form of repelling fixed point, since a system that starts near a saddle point will not stay near it, the dynamics being repelled along the unstable direction.

Unless we are very lucky, however, the Jacobian matrix is unlikely to be diagonal. In the general case it will have off-diagonal as well as diagonal elements and the solution above will not be correct. With a little more work, however, we can make progress in this case too. The trick is to find combinations of the variables x and y that move independently as x and y alone do above.

Consider the combinations of variables

$$\xi_1 = a\epsilon_x + b\epsilon_y, \qquad \xi_2 = c\epsilon_x + d\epsilon_y.$$

(18.26)

In matrix form we can write these as

$$\begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \epsilon_x \\ \epsilon_y \end{pmatrix},$$

(18.27)

or simply

$$\boldsymbol{\xi} = \mathbf{Q}\boldsymbol{\epsilon},$$

(18.28)

where **Q** is the matrix of the coefficients a, b, c, d.

The time evolution of $\boldsymbol{\xi}$ close to the fixed point is given by

$$\frac{d\boldsymbol{\xi}}{dt} = \mathbf{Q}\frac{d\boldsymbol{\epsilon}}{dt} = \mathbf{Q}\mathbf{J}\boldsymbol{\epsilon} = \mathbf{Q}\mathbf{J}\mathbf{Q}^{-1}\boldsymbol{\xi}$$

(18.29)

where we have used Eqs. (18.19) and (18.28). If ξ_1 and ξ_2 are to evolve independently, then we require that the matrix $\mathbf{QJQ}^{-1}$ be diagonal, just as **J** itself was in the simple case we studied above. Linear algebra then tells us that **Q** must be the matrix of eigenvectors of **J**. More specifically, since **J** is in general asymmetric, **Q** is the matrix whose rows are the left eigenvectors of **J** and $\mathbf{Q}^{-1}$ is the inverse of that matrix, which is the matrix whose columns are the right eigenvectors of **J** (since the left and right eigenvectors of a matrix are mutually orthogonal).

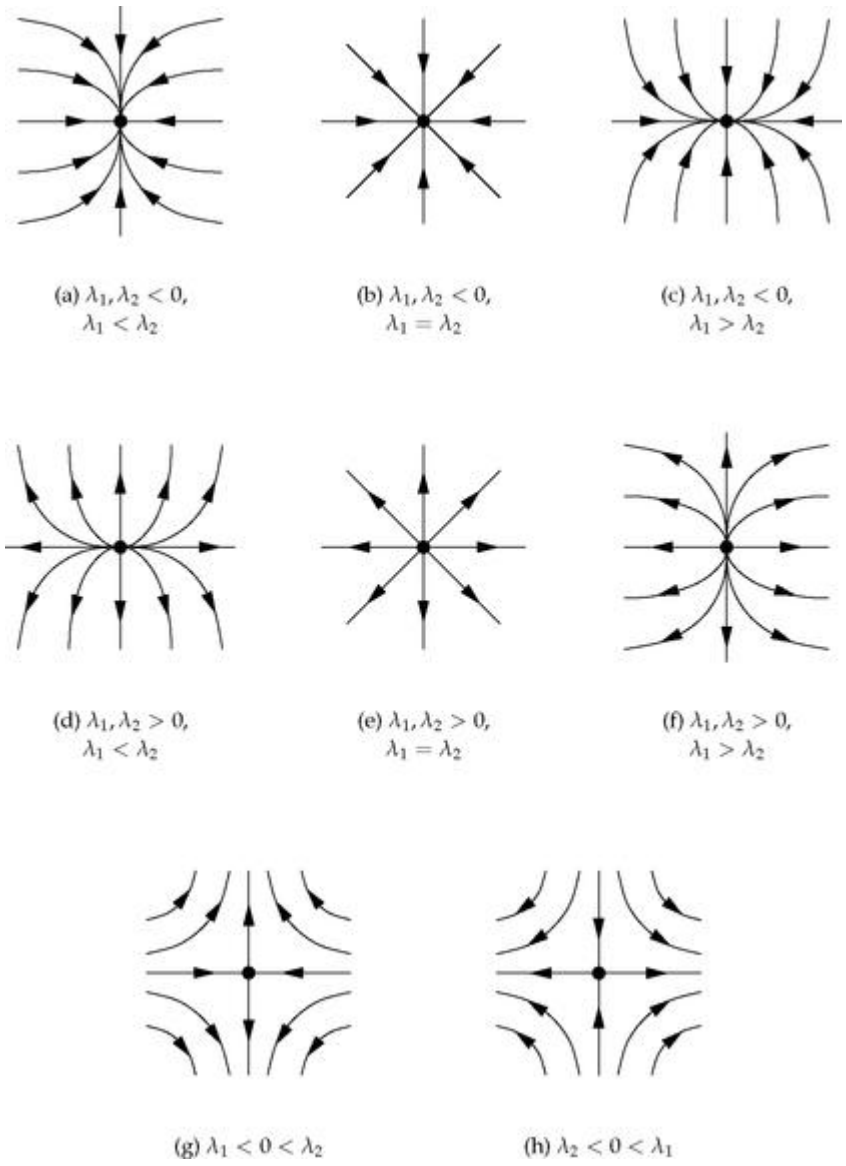


Figure 18.1: Flows in the vicinity of different types of fixed points. The flows around a fixed point in a two-variable dynamical system with a diagonal Jacobian matrix, as described in the text, can take a variety of different forms as shown. (a), (b), and (c) are all attracting fixed points, (d), (e), and (f) are repelling, and (g) and (h) are saddle points.

Thus, provided we can find the eigenvectors of **J** we can also find the combinations ξ_1 and ξ_2 that move independently of one another near the fixed point. These combinations satisfy the equations

$$\frac{d\xi_1}{dt} = \lambda_1 \xi_1, \qquad \frac{d\xi_2}{dt} = \lambda_2 \xi_2,$$

(18.30)

where λ_1 and λ_2 are the elements of our diagonal matrix, which are also the eigenvalues of **J** corresponding to the two eigenvectors. Equation (18.30) has the obvious solution

$$\xi_1(t) = \xi_1(0)\,e^{\lambda_1 t}, \qquad \xi_2(t) = \xi_2(0)\,e^{\lambda_2 t}.$$

(18.31)

The lines $\xi_1 = 0$ and $\xi_2 = 0$ play the role of the axes in Fig. 18.1—they are lines along which we move either directly away from or directly towards the fixed point—and Eq. (18.31) indicates that our distance from the fixed point along these lines will either grow or decay exponentially according to the signs of the two eigenvalues. Since the eigenvectors of an asymmetric matrix are not in general orthogonal to one another, these lines are not in general at right angles, so the flows around the fixed point will look similar to those of Fig. 18.1 but squashed, as shown in Fig. 18.2. Nonetheless, we can still classify our fixed points as attracting, repelling, or saddle points as shown in the figure. Similar analyses can be performed for systems with larger numbers of variables and the basic results are the same: by finding the eigenvectors of the Jacobian matrix we can determine the combinations of variables that move independently and hence solve the evolution of the system in the vicinity of the fixed point.

There is another subtlety that arises for systems of two or more variables that is not found in the one-variable case. The eigenvalues of an asymmetric matrix need not be real. Even if the elements of the matrix itself are real, the eigenvalues can be imaginary or complex. What does it mean if the eigenvalues of the Jacobian matrix in our derivation are complex? Putting such eigenvalues into Eq. (18.31) gives us a solution that *oscillates* around the fixed point, rather than simply growing or decaying. Indeed, the substitution actually gives us a value for ξ_1 and ξ_2 that itself is complex, which looks like it might be a problem, since the coordinates are supposed to be real. However, our equations are linear, so the real part of that solution is also a solution, as is the imaginary part, or any combination of the two.

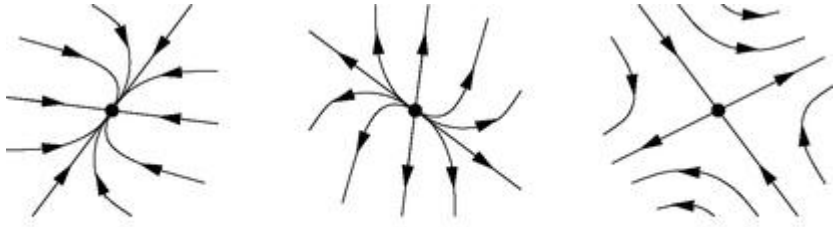


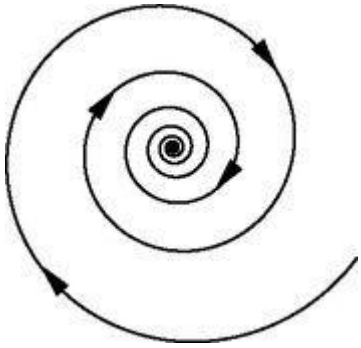
Figure 18.2: Examples of flows around general fixed points. When the Jacobian matrix is not diagonal the flows around a fixed point look like squashed or stretched versions of those in Fig. 18.1.

If $\lambda_1 = \alpha + i\omega$, for example, where α and ω are real numbers, then the general real solution for ξ_1 is

$$\xi_1(t) = \mathrm{Re}\left[C\,e^{(\alpha+i\omega)t}\right] = e^{\alpha t}\left(A\cos\omega t + B\sin\omega t\right),$$

(18.32)

where A and B are real constants and C is a complex constant. Thus the solution is the product of a part that oscillates and a part that either grows or decays exponentially. For the case of two variables, it turns out that the eigenvalues are always either both real or both complex, and if both are complex then they are complex conjugates of one another. In the latter case, both ξ_1 and ξ_2 then have this combined behavior of oscillation with exponential growth or decay, with the same frequency ω of oscillation and the same rate of growth or decay. The net result is a trajectory that describes a spiral around the fixed point. Depending on whether α is positive or negative the spiral either moves outward from the fixed point or inward. If it moves inward, i.e., if $\alpha < 0$. then the fixed point is a stable one; otherwise, of $\alpha > 0$, it is unstable. Thus stability is in this case determined solely by the real part of the eigenvalues. (In the special case where $\alpha = 0$ we must, as before, look at higher-order terms in the expansion around the fixed point to determine the nature of the point.)



The flows around a fixed point whose Jacobian matrix has complex eigenvalues describe a spiral.

When there are more than two variables, the eigenvalues must either be real or they appear in complex conjugate pairs. Thus again we have eigendirections that simply grow or decay, or that spiral in or out.

We are, however, not done yet. There is a further interesting behavior arising in systems with two or more variables that will be important when we come to study networked systems. In addition to fixed points, one also finds in some systems *limit cycles*. A limit cycle is a closed loop in the dynamics such that a system finding itself on such a loop remains there indefinitely, circulating around and returning repeatedly to its starting point. Limit cycles can be treated in many ways rather like fixed points: we can study the dynamics close to the limit cycle by expanding in a small displacement coordinate. Like fixed points, limit cycles tend to be either attracting or repelling, meaning that points close to them either spiral inwards toward the limit cycle or outwards away from it.

Physically, limit cycles represent stable oscillatory behaviors in systems. We mentioned one such behavior in Section 17.5 in our brief discussion of the SIRS model. In certain parameter regimes, the SIRS model can show "waves" of infection—oscillatory behaviors under which a disease infects a large fraction of the population, who then recover and gain immunity, reducing substantially the number of victims available to the disease and therefore causing the number of cases to drop dramatically. When the first wave of individuals later loses their immunity they move back into the susceptible state, become infected again, and another wave starts. Another example of oscillation in a dynamical system is the oscillation of the numbers of predators and prey in a two-species ecosystem represented, for example, by the Lotka-Volterra predator-prey

equations [307]. Such oscillations have been famously implicated in the mysterious periodic variation in the populations of hares and lynx recorded by the Hudson Bay Company in Canada during the nineteenth century. A further discussion of this and other aspects of limit cycles can be found in Ref. [307].

18.2 DYNAMICS ON NETWORKS

Let us now apply some of the ideas of the previous section to dynamical systems on networks. First, we need to be clear what we mean by such systems. Typically, we mean that we have independent dynamical variables x_i, y_i, . . . on each vertex i of our network and that they are coupled together only along the edges of the network. That is, when we write our equation for the time evolution of a variable x_i, the individual terms appearing in that equation each involve only x_i, other variables on vertex i, or one or more variables on a vertex adjacent to i in the network. There are no terms involving variables on non-adjacent vertices and no terms involving variables on more than one adjacent vertex.

An example of a dynamical system of this type is our equation (17.35) for the probability of infection of a vertex in the network version of the SI epidemic model:

$$\frac{\mathrm{d}x_i}{\mathrm{d}t} = \beta(1-x_i)\sum_j A_{ij}x_j.$$

(18.33)

This equation only has terms involving pairs of variables that are connected by edges since these are the only pairs for which A_{ij} is non-zero.

For a system with a single variable on each vertex we can write a general first-order equation

$$\frac{\mathrm{d}x_i}{\mathrm{d}t} = f_i(x_i) + \sum_j A_{ij}g_{ij}(x_i, x_j),$$

(18.34)

where we have separated terms that involve variables on adjacent vertices from those that do not. You can think of f_i as specifying the intrinsic dynamics of a vertex—it specifies how the variable x_i would evolve in the absence of any connections between vertices, i.e., if $A_{ij} = 0$ for all i, j. Conversely, g_{ij} describes the contribution from the connections themselves; it represents the coupling between variables on different vertices.

Notice that we have specified different functions f_i and g_{ij} for each vertex or pair of vertices, so the dynamics obeyed by each vertex can be different. In many cases, however, when each of the vertices represents a similar thing—such as a person in the case of an epidemic model—the dynamics for each vertex may be the same, or at least similar enough that we can ignore any differences. In such cases, the functions in Eq. (18.34) are the same for all vertices and the equation becomes

$$\frac{dx_i}{dt} = f(x_i) + \sum_j A_{ij} g(x_i, x_j).$$

(18.35)

In the examples in this chapter we will assume that this is the case. We will also assume that the network is undirected so that A_{ij} is symmetric—if x_i is affected by x_j then x_j is similarly affected by x_i. (Note, however, that we do not assume that the function g is symmetric in its arguments: $g(u, v) \neq g(v, u)$.) Again, the SI model of Eq. (18.33) is an example of a system of this kind, one in which $f(x) = 0$ and $g(x_i, x_j) = \beta(1 - x_i)x_j$.

18.2.1 LINEAR STABILITY ANALYSIS

Let us try applying the tools of linear stability analysis to Eq. (18.35). Suppose we are able to find a fixed point $\{x_i^*\}$ of Eq. (18.35) by solving the simultaneous equations

$$f(x_i^*) + \sum_j A_{ij} g(x_i^*, x_j^*) = 0$$

(18.36)

for all *i*. Note that finding a fixed point in this case means finding a value $x_i = x_i^*$ for every vertex *i*—the fixed point is the complete set $\{x_i^*\}$. Note also that in general the position of the fixed point depends both on the particular dynamical process taking place on the network (via the functions *f* and *g*) and on the structure of the network (via the adjacency matrix). If either is changed then the position of the fixed point will also change.

Now we can linearize about this fixed point in the usual way by writing $x_i = x_i^* + \epsilon_i$, performing a multiple Taylor expansion in all variables simultaneously, and dropping terms at second order in small quantities and higher:

$$\begin{aligned}
\frac{\mathrm{d}x_i}{\mathrm{d}t} = \frac{\mathrm{d}\epsilon_i}{\mathrm{d}t} &= f(x_i^* + \epsilon_i) + \sum_j A_{ij} g(x_i^* + \epsilon_i, x_j^* + \epsilon_j) \\
&= f(x_i^*) + \epsilon_i \frac{\mathrm{d}f}{\mathrm{d}x}\bigg|_{x=x_i^*} + \sum_j A_{ij} g(x_i^*, x_j^*) \\
&\qquad + \epsilon_i \sum_j A_{ij} \frac{\partial g(u,v)}{\partial u}\bigg|_{u=x_i^*, v=x_j^*} + \sum_j A_{ij}\epsilon_j \frac{\partial g(u,v)}{\partial v}\bigg|_{u=x_i^*, v=x_j^*} + \cdots \\
&= \epsilon_i \frac{\mathrm{d}f}{\mathrm{d}x}\bigg|_{x=x_i^*} + \epsilon_i \sum_j A_{ij} \frac{\partial g(u,v)}{\partial u}\bigg|_{u=x_i^*, v=x_j^*} + \sum_j A_{ij}\epsilon_j \frac{\partial g(u,v)}{\partial v}\bigg|_{u=x_i^*, v=x_j^*} + \cdots,
\end{aligned}$$

(18.37)

where we have used Eq. (18.36).

If we know the position of the fixed point, then the derivatives in these expressions are simply numbers. For convenience, let us write

$$\alpha_i = \frac{\partial f}{\partial x}\bigg|_{x=x_i^*},$$

(18.38a)

$$\beta_{ij} = \left.\frac{\partial g(u,v)}{\partial u}\right|_{u=x_i^*, v=x_j^*},$$

(18.38b)

$$\gamma_{ij} = \left.\frac{\partial g(u,v)}{\partial v}\right|_{u=x_i^*, v=x_j^*}.$$

(18.38c)

Then

$$\frac{\mathrm{d}\epsilon_i}{\mathrm{d}t} = \left[\alpha_i + \sum_j \beta_{ij} A_{ij}\right]\epsilon_i + \sum_j \gamma_{ij} A_{ij}\epsilon_j,$$

(18.39)

which we can write in matrix form as

$$\frac{\mathrm{d}\boldsymbol{\epsilon}}{\mathrm{d}t} = \mathbf{M}\boldsymbol{\epsilon},$$

(18.40)

where **M** is the matrix with elements

$$M_{ij} = \delta_{ij}\left[\alpha_i + \sum_k \beta_{ik} A_{ik}\right] + \gamma_{ij} A_{ij},$$

(18.41)

and δ_{ij} is the Kronecker delta.

We can solve Eq. (18.40) by writing $\in$ as a linear combination of the eigenvectors of **M**, specifically the *right* eigenvectors, since **M** is in general not symmetric:

$$\epsilon(t) = \sum_r c_r(t)\mathbf{v}_r\,,$$

(18.42)

so that Eq. (18.40) becomes

$$\sum_r \frac{\mathrm{d}c_r}{\mathrm{d}t}\mathbf{v}_r = \mathbf{M}\sum_r c_r(t)\mathbf{v}_r = \sum_r \mu_r c_r(t)\mathbf{v}_r\,,$$

(18.43)

where μ_r is the eigenvalue corresponding to the eigenvector $\mathbf{v}_r$. Comparing terms in each eigenvector we then have

$$\frac{\mathrm{d}c_r}{\mathrm{d}t} = \mu_r c_r(t),$$

(18.44)

which implies that

$$c_r(t) = c_r(0)\,\mathrm{e}^{\mu_r t}.$$

(18.45)

Immediately we see that if the real parts of all of the eigenvalues μ_r are negative, then $c_r(t)$—and hence $\in$—is decaying in time for all r and our fixed point will be attracting. If the real parts are all positive the fixed point will be repelling. And if some are positive and some are negative then the fixed point is a saddle, although, as before, this is perhaps best looked at as a form of repelling fixed point: the flows near a saddle have at least one repelling direction, which means that a system starting in the vicinity of such a point will not in general stay near it, regardless of whether the other directions are attracting or not.

18.2.2 SPECIAL CASES

Let us look at some common special cases of the general formalism above. A particularly simple case is when the fixed point is *symmetric*, meaning that x_i^* has the same value for every i: $x_i^* = x^*$. This occurs in the SI model for instance—there is a fixed point at $x_i^* = 1$ for all i.

For a symmetric fixed point, the fixed point equation, Eq. (18.36), becomes

$$f(x^*) + \sum_j A_{ij} g(x^*, x^*) = f(x^*) + k_i g(x^*, x^*) = 0,$$

(18.46)

where k_i is the degree of vertex i and we have made use of $k_i = \sum_j A_{ij}$ (see Eq. (6.19)). Given the appearance of k_i here, there are only two ways this equation can be satisfied for all i: either all vertices must have the same degree or $g(x^*, x^*) = 0$. Since the former is not really realistic—few networks of interest have all degrees the same—let us concentrate on the latter and assume that

$$g(x^*, x^*) = 0.$$

(18.47)

Again the SI model provides an example of this type of behavior. The coupling function g is of the form $\beta x(1 - x)$ in that model, which is zero at the two fixed points at $x = 0, 1$.

Equations (18.46) and (18.47) together imply also that $f(x^*) = 0$ and hence the fixed point x^* is the same in this case as the fixed point for the "intrinsic" dynamics of a vertex: it falls at the same place as it would if there were no connections between vertices at all. The position of the fixed point is also independent of the network structure in this case, a point that will shortly be important.

For a symmetric fixed point, the quantities α_i, β_{ij}, and γ_{ij} defined in Eq. (18.38) become

$$\alpha_i = \alpha = \left.\frac{\partial f}{\partial x}\right|_{x=x^*},$$

(18.48a)

$$\beta_{ij} = \beta = \left.\frac{\partial g(u,v)}{\partial u}\right|_{u,v=x^*},$$

(18.48b)

$$\gamma_{ij} = \gamma = \left.\frac{\partial g(u,v)}{\partial v}\right|_{u,v=x^*}.$$

(18.48c)

Then Eq. (18.39) becomes

$$\frac{d\epsilon_i}{dt} = (\alpha + \beta k_i)\epsilon_i + \gamma \sum_j A_{ij}\epsilon_j.$$

(18.49)

The situation simplifies further if the coupling function $g(x_i, x_j)$ depends only on x_j and not on x_i, i.e., if x_i obeys an equation of the form $dx_i/dt = f(x_i) + \sum_j A_{ij} g(x_j)$. Then $\beta = 0$ and

$$\frac{d\epsilon_i}{dt} = \alpha\epsilon_i + \gamma \sum_j A_{ij}\epsilon_j,$$

(18.50)

which we can write in matrix form as

$$\frac{d\boldsymbol{\epsilon}}{dt} = (\alpha\mathbf{I} + \gamma\mathbf{A})\boldsymbol{\epsilon}.$$

(18.51)

As in the general case, the fixed point will be stable if and only if all of the eigenvalues of the matrix $\alpha\mathbf{I} + \gamma\mathbf{A}$ are negative. Let $\mathbf{v}_r$ be the eigenvector of the adjacency matrix with eigenvalue κ_r.

Then

$$(\alpha\mathbf{I} + \gamma\mathbf{A})\mathbf{v}_r = \alpha\mathbf{I}\mathbf{v}_r + \gamma\mathbf{A}\mathbf{v}_r = \alpha\mathbf{v}_r + \gamma\kappa_r\mathbf{v}_r = (\alpha + \gamma\kappa_r)\mathbf{v}_r.$$

(18.52)

Hence $\mathbf{v}_r$ is also an eigenvector of $\alpha\mathbf{I} + \gamma\mathbf{A}$, but with eigenvalue $\alpha + \gamma\kappa_r$. Now if all eigenvalues are to be negative, we require that

$$\alpha + \gamma\kappa_r < 0$$

(18.53)

for all r and from this we can deduce a number of things. First of all it implies that $\alpha < -\gamma\kappa r$ for all r. The adjacency matrix always has both positive and negative eigenvalues (a result that we will prove in Section 18.3.2), which means that for this inequality to be satisfied for all r we must have $\alpha < 0$. If $\alpha > 0$ then the fixed point is never stable.

Second, we can rearrange Eq. (18.53) to give

$$\kappa_r < -\alpha/\gamma \qquad \text{if } \gamma > 0,$$

(18.54a)

$$\kappa_r > -\alpha/\gamma \qquad \text{if } \gamma < 0,$$

(18.54b)

for all r. Note, however, that if Eq. (18.54a) is satisfied for the largest (most positive) eigenvalue κ_1 of the adjacency matrix, then it is necessarily satisfied by all the other eigenvalues as well. Similarly if Eq. (18.54b) is satisfied for the most negative eigenvalue κ_n then it is satisfied by all others. Thus the conditions above can be simplified to a single condition each:

$$\kappa_1 < -\alpha/\gamma \qquad \text{if } \gamma > 0,$$

(18.55a)

$$\kappa_n > -\alpha/\gamma \qquad \text{if } \gamma < 0.$$

(18.55b)

Alternatively, we can take reciprocals of these conditions and combine them into a single statement:

$$\frac{1}{\kappa_n} < -\frac{\gamma}{\alpha} < \frac{1}{\kappa_1},$$

(18.56)

If we want we can fill in the explicit values of α and γ thus:

$$\frac{1}{\kappa_n} < -\left[\frac{\mathrm{d}g}{\mathrm{d}x}\Big/\frac{\mathrm{d}f}{\mathrm{d}x}\right]_{x=x^*} < \frac{1}{\kappa_1},$$

(18.57)

where we have written g as a function of a single variable since, by hypothesis, it only depends on one argument in this case.

Equation (18.57) is sometimes called a *master stability condition*. It has a special form: note that κ_1 and κ_n depend only on the structure of the network and not on anything about the dynamics, while α and γ depend only the nature of the dynamics and not on the network structure. Thus Eq. (18.57) effectively gives us a single condition that must be satisfied by any type of dynamics and its associated fixed point if that dynamics is to be stable on our network. Or conversely, it gives a condition on the network structure, via the largest and smallest eigenvalues, that guarantees stability of a given fixed point for a given type of dynamics.

Another case where we can derive a master stability condition is the case in which the coupling function g depends on its two arguments according to $g(x_i, x_j) = g(x_i) - g(x_j)$. A physicist might think of this as a "spring-like" interaction—if $g(x)$ were a simple linear function of its argument then x_i and x_j would act upon one another like two masses coupled by a spring, exerting forces that depend on the difference of their positions. More generally, $g(x)$ is non-linear and we have a non-linear spring.

For this choice of coupling, and still assuming a symmetric fixed point, we have $g(x^*, x^*) = 0$ as before and hence also $f(x^*) = 0$, and the quantities defined in Eq. (18.38) become

$$\alpha_i = \alpha = \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_{x=x^*},$$

(18.58a)

$$\beta_{ij} = \beta = \left.\frac{dg}{dx}\right|_{x=x^*},$$

(18.58b)

$$\gamma_{ij} = -\beta.$$

(18.58c)

Then Eq. (18.39) becomes

$$\begin{aligned}\frac{d\epsilon_i}{dt} &= (\alpha + \beta k_i)\epsilon_i - \beta \sum_j A_{ij}\epsilon_j \\ &= \alpha\epsilon_i + \beta \sum_j (k_i\delta_{ij} - A_{ij})\epsilon_j,\end{aligned}$$

(18.59)

or in matrix form

$$\frac{d\boldsymbol{\epsilon}}{dt} = (\alpha\mathbf{I} + \beta\mathbf{L})\boldsymbol{\epsilon},$$

(18.60)

where **L** is the matrix with elements

$$L_{ij} = k_i\delta_{ij} - A_{ij}.$$

(18.61)

We have encountered this matrix before. It is the *graph Laplacian*—see Eq. (6.43).

Equation (18.60) is of the same form as Eq. (18.51), with the adjacency matrix replaced by the graph Laplacian. Thus we can immediately see that the fixed point will be stable if and only if the eigenvalues λ_r of the Laplacian satisfy

$$\alpha + \beta\lambda_r < 0$$

(18.62)

for all r.

As shown in Section 6.13.2, the smallest eigenvalue of the Laplacian matrix is always zero, and hence Eq. (18.62), when applied to the smallest eigenvalue, implies again that $\alpha < 0$ is a necessary (but not sufficient) condition for the fixed point to be stable, or equivalently

$$\left.\frac{df}{dx}\right|_{x=x^*} < 0.$$

(18.63)

Assuming this condition is satisfied then, since all eigenvalues of the Laplacian are non-negative it follows that $1/\lambda_r > -\beta/\alpha$ for stability, regardless of the sign of β. Furthermore, if this condition is true for the largest eigenvalue, traditionally denoted λ_n, then it is true for all smaller eigenvalues as well, so the requirement for stability can be reduced to the requirement that $1/\lambda_n > -\beta/\alpha$, or

$$\frac{1}{\lambda_n} > -\left[\frac{dg}{dx}\Big/\frac{df}{dx}\right]_{x=x^*},$$

(18.64)

along with the condition in Eq. (18.63).

Again, Eq. (18.64) neatly separates questions of dynamics from questions of network structure. The structure appears only on the left of the inequality, via the eigenvalues of the graph Laplacian, and the dynamics appears only on the right, via derivatives of the functions f and g.

Apart from establishing a condition for the stability of a fixed point, the master stability condition is of particular interest in the study of bifurcations—situations in which a fixed point loses stability as the parameters of a system change. If we vary parameters appearing in the definitions of f and g, for example, then we can cause a fixed point that initially satisfies a condition like (18.64) to stop satisfying it and so become unstable. In practice, this means that the system will suddenly change its behavior as it passes through the point where $1/\lambda_n = -\beta/\alpha$. At one moment it will be sitting happily at its stable fixed point, going nowhere, and at the next, as that point becomes unstable, it will start moving, gathering speed exponentially, and quite likely wind up in some completely different state far from where it started, as it falls into the basin of attraction

of a different stable fixed point or limit cycle. We will see some examples of behavior of this kind shortly.

18.2.3 AN EXAMPLE

As an example, consider the following simple model of "gossip," or diffusion of an idea or fad across a social network. Suppose some new idea is circulating through a community and x_i represents the amount person i is talking about it, which will be governed by an equation of the form (18.35). We will put

$$f(x) = a(1 - x)$$

(18.65)

with $a > 0$, which means that the intrinsic dynamics of a single vertex has a stable fixed point at $x^* = 1$—each person has an intrinsic tendency to talk this much about the latest craze, regardless of whether their friends want to hear about it or not. For the interaction term we will assume that people tend to copy their friends: they increase the amount they are talking about whatever it is if their friends are talking about it more than they are, and decrease if their friends are talking about it less. We represent this by putting $g(x_i, x_j) = g(x_j) - g(x_i)$ with

$$g(x) = \frac{bx}{1 + x}$$

(18.66)

and $b > 0$. This is an increasing function of its argument, as it should be, but saturates when $x \gg 1$—beyond some point, it makes no difference if your friends shout louder.

Now we can apply the general formalism developed above. The symmetric fixed point for the model is at $x_i = 1$ for all i. At this point everyone is talking about the topic *du jour* with equal enthusiasm. This fixed point, however, is stable only provided Eqs. (18.63) and (18.64) are satisfied. Equation (18.63) is always satisfied, given that $a > 0$. Equation 18.64 implies that $1/\lambda_n > b/4a$, or equivalently

$$\lambda_n < \frac{4a}{b}.$$

(18.67)

Thus we can make the fixed point unstable, for example, by increasing b to the point where the

right-hand side of this inequality falls below the largest eigenvalue λ_n of the Laplacian for the particular network we are looking at. Increasing b in this case corresponds to increasing the amount of influence your friends have on you.

And what happens when the fixed point becomes unstable? There are no other symmetric fixed points for this particular system, since there are no other values that give $f(x) = 0$ (which is a requirement for our symmetric fixed point). So the system cannot switch to another symmetric fixed point. One possibility is that the variables might diverge to $\pm\infty$, and this happens in some systems, but not in this one, where the form of $f(x)$ prevents it. Another possibility is that the system might begin to oscillate, or even enter a chaotic regime in which it meanders around in pseudorandom fashion indefinitely. In the present case, however, it does something simpler. It moves to a *non-symmetric* fixed point, one in which the fixed-point values of the variables x_i are not all equal. This is an interesting and perhaps unexpected behavior. Our calculations are telling us when the influence between neighboring individuals in the network becomes very strong that instead of driving everyone to behave in the same way, as one might expect, it actually causes behaviors to differ. People spontaneously develop idiosyncrasies and start doing things their own way.

18.3 DYNAMICS WITH MORE THAN ONE VARIABLE PER VERTEX

Our developments so far have assumed that there is only a single variable x_i on each vertex i of the network. Many systems, however, have more than one variable per vertex. The epidemiological examples of Chapter 17, mostly have several—s, x, r, and so forth.

Consider a system with an arbitrary number of variables $x_1^i, x_2^i, \ldots$ on each vertex i, but let us assume that we have the same number of variables on each vertex and that, as before, they obey equations of the same form. For convenience let us write the set of variables on a single vertex as a vector $\mathbf{x}^i = (x_1^i, x_2^i, \ldots)$ and then write the equations governing their time evolution as

$$\frac{\mathrm{d}\mathbf{x}^i}{\mathrm{d}t} = \mathbf{f}(\mathbf{x}^i) + \sum_j A_{ij}\mathbf{g}(\mathbf{x}^i, \mathbf{x}^j).$$

(18.68)

Note that the functions f and g, representing the intrinsic dynamics and the coupling, have now become vector functions **f** and **g** of vector arguments, with the same rank as **x**.

Following the same line of reasoning as before, we can study the stability of a symmetric fixed point $\mathbf{x}^i = \mathbf{x}^*$ by writing $\mathbf{x}^i = \mathbf{x}^* + \in^i$ and performing a Taylor expansion. The resulting linearized equation for the evolution of the μth component of $\in^i$ is then

$$\begin{aligned}\frac{\mathrm{d}\epsilon_\mu^i}{\mathrm{d}t} &= \left[\epsilon_1^i \frac{\partial f_\mu(\mathbf{x})}{\partial x_1} + \epsilon_2^i \frac{\partial f_\mu(\mathbf{x})}{\partial x_2} + \ldots\right]_{\mathbf{x}=\mathbf{x}^*} \\ &\quad + \sum_j A_{ij}\left[\epsilon_1^i \frac{\partial g_\mu(\mathbf{u},\mathbf{v})}{\partial u_1} + \epsilon_2^i \frac{\partial g_\mu(\mathbf{u},\mathbf{v})}{\partial u_2} + \ldots + \epsilon_1^j \frac{\partial g_\mu(\mathbf{u},\mathbf{v})}{\partial v_1} + \epsilon_2^j \frac{\partial g_\mu(\mathbf{u},\mathbf{v})}{\partial v_2} + \ldots\right]_{\mathbf{u},\mathbf{v}=\mathbf{x}^*} \\ &= \sum_\nu \left[\epsilon_\nu^i \frac{\partial f_\mu(\mathbf{x})}{\partial x_\nu}\bigg|_{\mathbf{x}=\mathbf{x}^*} + k_i \epsilon_\nu^i \frac{\partial g_\mu(\mathbf{u},\mathbf{v})}{\partial u_\nu}\bigg|_{\mathbf{u},\mathbf{v}=\mathbf{x}^*} + \sum_j A_{ij}\epsilon_\nu^j \frac{\partial g_\mu(\mathbf{u},\mathbf{v})}{\partial v_\nu}\bigg|_{\mathbf{u},\mathbf{v}=\mathbf{x}^*}\right],\end{aligned}$$

(18.69)

where f_μ and g_μ represent the μth components of **f** and **g**.

As before, the derivatives in this expression are simply constants, and for convenience let us define

$$\alpha_{\mu\nu} = \frac{\partial f_\mu(\mathbf{x})}{\partial x_\nu}\bigg|_{\mathbf{x}=\mathbf{x}^*},$$

(18.70a)

$$\beta_{\mu\nu} = \left.\frac{\partial g_\mu(\mathbf{u},\mathbf{v})}{\partial u_\nu}\right|_{\mathbf{u},\mathbf{v}=\mathbf{x}^*},$$

(18.70b)

$$\gamma_{\mu\nu} = \left.\frac{\partial g_\mu(\mathbf{u},\mathbf{v})}{\partial v_\nu}\right|_{\mathbf{u},\mathbf{v}=\mathbf{x}^*},$$

(18.70c)

so that

$$\begin{aligned}\frac{\mathrm{d}\epsilon_\mu^i}{\mathrm{d}t} &= \sum_\nu\left[(\alpha_{\mu\nu} + k_i\beta_{\mu\nu})\epsilon_\nu^i + \sum_j A_{ij}\gamma_{\mu\nu}\epsilon_\nu^j\right] \\ &= \sum_{j\nu}\left[\delta_{ij}(\alpha_{\mu\nu} + k_i\beta_{\mu\nu}) + A_{ij}\gamma_{\mu\nu}\right]\epsilon_\nu^j\end{aligned}$$

(18.71)

where δ_{ij} is the Kronecker delta again.
We can write this equation in the matrix form

$$\frac{\mathrm{d}\boldsymbol{\epsilon}}{\mathrm{d}t} = \mathbf{M}\boldsymbol{\epsilon},$$

(18.72)

where **M** is a matrix whose rows (and columns) are labeled by a double pair of indices (i, μ) and whose elements are

$$M_{i\mu,j\nu} = \delta_{ij}\alpha_{\mu\nu} + \delta_{ij}k_i\beta_{\mu\nu} + A_{ij}\gamma_{\mu\nu}.$$

(18.73)

In principle, we can now determine whether the fixed point is stable by examining the eigenvalues of this new matrix. If the real parts of the eigenvalues are all negative then the fixed point is stable, otherwise it is not. In practice this can be a difficult thing to do in general but, as before, there are some common special cases where the calculation simplifies, yielding a master stability condition.

18.3.1 SPECIAL CASES

As before we consider the case where $\mathbf{g}(\mathbf{x}^i, \mathbf{x}^j)$ depends only on its second argument and not on its first. In this case $\beta_{\mu\nu} = 0$ for all μ, ν and Eq. (18.71) becomes

$$\frac{\mathrm{d}\epsilon_\mu^i}{\mathrm{d}t} = \sum_{j\nu}\left[\delta_{ij}\alpha_{\mu\nu} + A_{ij}\gamma_{\mu\nu}\right]\epsilon_\nu^j.$$

(18.74)

Now let v_r^i be the ith component of the eigenvector $\mathbf{v}_r$ of the adjacency matrix corresponding to eigenvalue κ_r. Let us write

$$\epsilon_\mu^i(t) = \sum_r c_\mu^r(t) v_r^i.$$

(18.75)

This equation expresses the vector of elements ϵ_μ^i as a linear combination of eigenvectors in the usual way, but with a separate set of coefficients c_μ^r for each dynamical variable μ. Substituting into Eq. (18.74), we get

$$\begin{aligned}\sum_r \frac{\mathrm{d}c_\mu^r}{\mathrm{d}t} v_r^i &= \sum_r \sum_{j\nu}\left[\delta_{ij}\alpha_{\mu\nu} + A_{ij}\gamma_{\mu\nu}\right] c_\nu^r(t) v_r^j \\ &= \sum_{r\nu}\left[\alpha_{\mu\nu} + \kappa_r\gamma_{\mu\nu}\right] c_\nu^r(t) v_r^i.\end{aligned}$$

(18.76)

Equating terms in the individual eigenvectors on both sides of the equation, we thus conclude that

$$\frac{\mathrm{d}c_\mu^r}{\mathrm{d}t} = \sum_\nu\left[\alpha_{\mu\nu} + \kappa_r\gamma_{\mu\nu}\right] c_\nu^r(t).$$

(18.77)

We can think of this as itself a matrix equation for a vector $\mathbf{c}^r = (c_1^r, c_2^r, \ldots)$ thus:

$$\frac{d\mathbf{c}^r}{dt} = [\boldsymbol{\alpha} + \kappa_r \boldsymbol{\gamma}]\mathbf{c}^r(t),$$

(18.78)

where $\boldsymbol{\alpha}$ and $\boldsymbol{\gamma}$ are matrices with elements $\alpha_{\mu\nu}$ and $\gamma_{\mu\nu}$, respectively.

This equation expresses the dynamics of the system close to the fixed point as a decoupled set of n separate systems, one for each eigenvalue κ_r of the adjacency matrix. If the fixed point of the system as a whole is to be stable, then each of these individual systems also needs to be stable, meaning that *their* eigenvalues need to be negative, or, more simply, the largest (i.e., most positive) eigenvalue of $\boldsymbol{\alpha} + \kappa_r\boldsymbol{\gamma}$ needs to be negative for every r.

Let us define the function $\sigma(\kappa)$ to be equal to the most positive eigenvalue of the matrix $\boldsymbol{\alpha} + \kappa\boldsymbol{\gamma}$, or the most positive real part in the case where the eigenvalues are complex. Typically this is an easy function to evaluate numerically. Notice that $\boldsymbol{\alpha} + \kappa\boldsymbol{\gamma}$ has only as many rows and columns as there are variables on each vertex of the network. If we have three variables on each vertex, for instance, the matrix has size 3×3, which is easily diagonalized.

The function $\sigma(\kappa)$ is called a *master stability function*. If our system is to be stable, the master stability function evaluated at the eigenvalue κ_r should be negative for all r:

$$\sigma(\kappa_r) < 0.$$

(18.79)

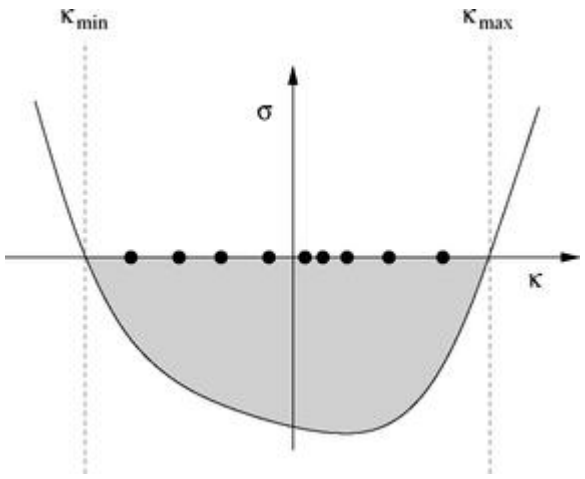


Figure 18.3: A sketch of a master stability function. One possible form for the master stability

function $\sigma(\kappa)$ might be as shown here (solid curve), with positive values for large and small κ but negative values in the intermediate range between $\kappa_{\min}$ and $\kappa_{\max}$. If all the eigenvalues of the adjacency matrix (represented by the dots) fall in this intermediate range, then the system is stable.

One possible form for the master stability function is shown in Fig. 18.3—it becomes large and positive for κ sufficiently small or sufficiently big, but is negative in some intermediate range $\kappa_{\min} < \kappa < \kappa_{\max}$. In that case, the system is stable provided all eigenvalues κ_r of the adjacency matrix fall in this range. Again this gives us a master stability condition that separates network structure from dynamics. The eigenvalues κ_r are properties solely of the structure, being derived from the adjacency matrix alone, while the limits $\kappa_{\min}$ and $\kappa_{\max}$ are properties solely of the dynamics, being derived from the matrices $\boldsymbol{\alpha}$ and $\boldsymbol{\gamma}$, which are determined by the derivatives of the functions f and g.

We can similarly write down the generalization of Eq. (18.58) to the case of many variables per vertex. If the interaction between vertices takes the form $\mathbf{g}(\mathbf{x}^i, \mathbf{x}^j) = \mathbf{g}(\mathbf{x}^i) - \mathbf{g}(\mathbf{x}^j)$, then $\gamma_{\mu\nu} = -\beta_{\mu\nu}$ and

$$\frac{\mathrm{d}\epsilon_\mu^i}{\mathrm{d}t} = \sum_{j\nu} \bigl[\delta_{ij}\alpha_{\mu\nu} + L_{ij}\beta_{\mu\nu}\bigr]\epsilon_\nu^j,$$

(18.80)

where $L_{ij} = \delta_{ij} - A_{ij}$ is an element of the Laplacian. Then the equivalent of Eq. (18.78) is

$$\frac{\mathrm{d}\mathbf{c}^r}{\mathrm{d}t} = [\boldsymbol{\alpha} + \lambda_r\boldsymbol{\beta}]\mathbf{c}^r(t),$$

(18.81)

where λ_r is an eigenvalue of the Laplacian and $\boldsymbol{\beta}$ is the matrix with elements $\beta_{\mu\nu}$. Again we can define a master stability function $\sigma(\lambda)$ equal to the most positive eigenvalue of $\boldsymbol{\alpha} + \lambda\beta$ (or the most positive real part for complex eigenvalues) and for overall stability of the system this function must be negative when $\lambda = \lambda_r$ for all r:

$$\sigma(\lambda_r) < 0.$$

(18.82)

And once again, for suitable forms of the master stability function, this allows us to develop a stability criterion that separates structure from dynamics.

18.3.2 SPECTRA OF COMPLEX NETWORKS

The formalism of the previous section turns questions about the stability of dynamical systems on networks into questions about the eigenvalue spectra of matrices. Given the definition of the dynamics taking place on the vertices of a network we calculate the master stability function and then the stability or not of the system depends on whether the master stability function is negative when evaluated at each of the eigenvalues of the appropriate matrix, such as the adjacency matrix or graph Laplacian. In particular, when the master stability function takes a relatively simple form like that sketched in Fig. 18.3, so that stability requires only that the eigenvalues fall in some specified range, then it is enough to know the smallest (most negative) and largest (most positive) eigenvalues of the matrix to insure stability—if the smallest and largest fall in the given range then necessarily all the others do too.

A number of results are known about the spectra of networks, and in particular about the smallest and largest eigenvalues, which allow us to make quite general theoretical statements about stability. For the adjacency matrix, for example, we can derive limits on the eigenvalues as follows.

Let $\mathbf{x}$ be an arbitrary real vector of n elements, which we will write as a linear combination of the eigenvectors $\mathbf{v}_r$ of the adjacency matrix $\mathbf{A}$ thus:

$$\mathbf{x} = \sum_r c_r \mathbf{v}_r.$$

(18.83)

Then

$$\frac{\mathbf{x}^T\mathbf{A}\mathbf{x}}{\mathbf{x}^T\mathbf{x}} = \frac{\sum_s c_s \mathbf{v}_s^T \mathbf{A} \sum_r c_r \mathbf{v}_r}{\sum_s c_s \mathbf{v}_s^T \sum_r c_r \mathbf{v}_r} = \frac{\sum_{rs} c_s c_r \kappa_r \mathbf{v}_s^T \mathbf{v}_r}{\sum_{rs} c_s c_r \mathbf{v}_s^T \mathbf{v}_r} = \frac{\sum_r c_r^2 \kappa_r}{\sum_r c_r^2} \le \frac{\sum_r c_r^2 \kappa_1}{\sum_r c_r^2} = \kappa_1,$$

(18.84)

where, as before, κ_1 is the largest eigenvalue and we have made use of the fact that $\mathbf{v}_s^T\mathbf{v}_r = \delta_{rs}$. This inequality is correct for any choice of $\mathbf{x}$. Thus, for instance, if $\mathbf{x} = \mathbf{1} = (1, 1, 1, \ldots)$ then

$$\kappa_1 \ge \frac{\mathbf{1}^T\mathbf{A}\mathbf{1}}{\mathbf{1}^T\mathbf{1}} = \frac{2m}{n} = \langle k \rangle.$$

(18.85)

So the largest eigenvalue of the adjacency matrix is never less than the average degree of the network.

Alternatively, suppose that vertex v is the highest-degree vertex in the network, with degree $k_{\max}$, and let us choose the elements of $\mathbf{x}$ thus:

$$x_i = \begin{cases} \sqrt{k_{\max}} & \text{if } i = v, \\ 1 & \text{if } A_{iv} = 1, \\ 0 & \text{otherwise.} \end{cases}$$

(18.86)

Then

$$\sum_j A_{ij} x_j \geq \left\{ \begin{matrix} k_{\max} & \text{if } i = v \\ \sqrt{k_{\max}} & \text{if } A_{iv} = 1 \\ 0 & \text{otherwise} \end{matrix} \right\} = \sqrt{k_{\max}}\, x_i \,.$$

(18.87)

(This result is non-trivial and you may find it helpful to work through each of the three cases to convince yourself that it is indeed correct.)

Multiplying both sides of Eq. (18.87) by x_i and summing over i we now get $\mathbf{x}^T\mathbf{A}\mathbf{x} \geq \sqrt{k_{\max}}\,\mathbf{x}^T\mathbf{x}$ or, using Eq. (18.84),

$$\kappa_1 \geq \frac{\mathbf{x}^T\mathbf{A}\mathbf{x}}{\mathbf{x}^T\mathbf{x}} \geq \sqrt{k_{\max}}.$$

(18.88)

Thus the largest eigenvalue of the adjacency matrix is never less than the square root of the largest degree.

Equations (18.85) and (18.88) imply that if we increase either the average or the maximum degree in our network, we will eventually increase the maximum eigenvalue also. In a system with a master stability function like that depicted in Fig. 18.3, this will in the end cause the system to become unstable.

We can also derive similar results for the lowest (most negative) eigenvalue κ_n of the adjacency matrix. We have

$$\frac{\mathbf{x}^T\mathbf{A}\mathbf{x}}{\mathbf{x}^T\mathbf{x}} = \frac{\sum_r c_r^2 \kappa_r}{\sum_r c_r^2} \geq \frac{\sum_r c_r^2 \kappa_n}{\sum_r c_r^2} = \kappa_n$$

(18.89)

for any real vector **x**. So, for instance, if vertex v is again the highest-degree vertex in the network and we make the choice

$$x_i = \begin{cases} \sqrt{k_{\max}} & \text{if } i = v, \\ -1 & \text{if } A_{iv} = 1, \\ 0 & \text{otherwise,} \end{cases}$$

(18.90)

then, following the same approach as before, we find that

$$\kappa_n \leq -\sqrt{k_{\max}}.$$

(18.91)

Thus increasing the highest degree in the network can also make the system unstable by the alternative route of decreasing the lowest eigenvalue. Whichever eigenvalue passes out of the region of stability first will be the one that makes the system unstable.

(We note in passing that Eqs. (18.88) and (18.91) together also tell us that the adjacency matrix of an undirected network always has both positive and negative eigenvalues, unless the network has no edges in it at all, in which case all eigenvalues are zero. We used this result previously in Section 18.2.2.)

Other results for the eigenvalues of the adjacency matrix can be derived for specific models of networks. For example, Chung *et al.* [68] have shown for the configuration model that the expected value of the largest eigenvalue in the limit of large network size is

$$\kappa_1 = \frac{\langle k^2 \rangle}{\langle k \rangle}.$$

(18.92)

In many cases this gives values of κ_1 considerably above the limits set by Eqs. (18.85) and (18.88). On configuration model networks with power-law degree distributions, for instance, where □k^2□

formally diverges in the limit of large n, we expect that κ_1 will similarly diverge.

One can also derive results for eigenvalues of the Laplacian. The smallest eigenvalue of the Laplacian is simple—it is always zero. For large networks the largest eigenvalue λ_n can be shown to lie in the range [18]

$$k_{\max} \leq \lambda_n \leq 2k_{\max},$$

(18.93)

which appears to be a relatively large range but in fact tells us a lot, ensuring again that the largest eigenvalue is guaranteed to increase if the highest degree in the network increases sufficiently.

18.4 SYNCHRONIZATION

A topic closely related to the study of dynamical stability is the study of synchronization. Many systems of scientific interest can be modeled as oscillators of one sort or another. The flashing of fireflies, the ticking of clocks, the synchronized clapping of a large audience, and the pathologically synchronized firing of brain cells during an epileptic attack can all be modeled as networks of oscillators coupled in such a way that the coupling causes the oscillators to synchronize.

The periodic, synchronized oscillations of such an oscillator network correspond, in dynamical systems terms, to a limit cycle of the overall dynamics (see Section 18.1.1). Like fixed points, limit cycles can be stable or unstable, attracting or repelling, depending on whether small perturbations away from the periodic behavior tend to grow or decay over time. The mathematics of whether synchronized states are stable is very similar to that for fixed points. Again one starts with a system of equations of the form of Eq. (18.68) but now assumes a periodic limit-cycle solution, $\mathbf{x}_i(t) = \mathbf{s}(t)$ for all i. Perturbing around this solution one can linearize the equations and, depending on the particular form of the interaction between vertices, expand the linearized solution as a combination of the eigenvectors of an appropriate matrix, such as the adjacency matrix or Laplacian. The result is a set of n decoupled systems, each oscillating independently and each of which must be stable if the system as a whole is to be stable. One can define a master stability function $\sigma(\lambda)$ again, corresponding to the growth rate of perturbations away from the periodic solution, which in this context is known as a *Lyapunov exponent*, although it plays exactly the same role as the leading eigenvector in our earlier analysis. Once again this master stability function must be negative when evaluated at each of the eigenvalues λ of the appropriate matrix and this gives us a condition for stability of the synchronized state.

Many details of the network synchronization process and many special cases have been studied. For a comprehensive discussion, the interested reader is encouraged to consult the review by Arenas *et al.* [23].

PROBLEMS

18.1 Consider a dynamical system on a k-regular network (i.e., one in which every vertex has the same degree k) satisfying

$$\frac{dx_i}{dt} = f(x_i) + \sum_j A_{ij} g(x_i, x_j),$$

and in which the initial condition is uniform over vertices, so that $x_i(0) = x_0$ for all i.

a. Show that $x_i(t) = x(t)$ for all i where

$$\frac{dx}{dt} = f(x) + kg(x, x),$$

and hence that one has to solve only one equation to solve the dynamics.
b. Show that for stability around a fixed point at $x_i = x^*$ for all i we require that

$$k < -\frac{1}{f'(x^*)} \left[\left(\frac{\partial}{\partial u} + \frac{\partial}{\partial v} \right) g(u, v) \right]_{u=v=x^*}.$$

18.2 Consider a dynamical system on an undirected network, with one variable per vertex obeying

$$\frac{dx_i}{dt} = f(x_i) + \sum_j A_{ij} [g(x_i) - g(x_j)],$$

as in Section 18.2.2. Suppose that the system has a symmetric fixed point at $x_i = x^*$ for all i.

a. Show, using results given in this chapter, that the fixed point is always stable if the largest degree $k_{\max}$ in the network satisfies

$$\frac{1}{k_{\max}} > -2 \left[\frac{dg}{dx} \Big/ \frac{df}{dx} \right]_{x=x^*}.$$

b. Suppose that $f(x) = rx(1 - x)$ and $g(x) = ax^2$. Show that there are two symmetric fixed points for this system, but that one if them is always unstable.
c. Give a condition on the maximum degree in the network that will ensure the stability of the other fixed point.

18.3 The dynamical systems we have considered in this chapter have all been on undirected networks, but systems on directed networks are possible too. Consider a dynamical system on a directed network in which the sign of the interaction along an edge attached to a vertex depends on the direction of the edge, ingoing edges having positive sign and outgoing edges having negative sign. An example of such a system is a food web of predator-prey interactions, in which an ingoing edge indicates in-flow of energy to a predator from its prey and an outgoing edge indicates out-flow from a prey to its predator. Such a system can be represented by a dynamics of the form

$$\frac{\mathrm{d}x_i}{\mathrm{d}t} = f(x_i) + \sum_j (A_{ij} - A_{ji}) g(x_i, x_j),$$

where g is a symmetric function of its arguments: $g(u, v) = g(v, u)$.

a. Consider a system of this form in which the in- and out-degrees of every vertex are equal to the same constant k. Show that such a system has a symmetric fixed point $x_i^* = x^*$ for all i satisfying $f(x^*) = 0$.
b. Writing $x_i = x^* + \epsilon_i$ linearize around this fixed point to show that in the vicinity of the fixed point the vector $\boldsymbol{\epsilon} = (\epsilon_1, \epsilon_2, \ldots)$ satisfies

$$\frac{\mathrm{d}\boldsymbol{\epsilon}}{\mathrm{d}t} = (\alpha \mathbf{I} + \beta \mathbf{M})\boldsymbol{\epsilon},$$

where $\mathbf{M} = \mathbf{A} - \mathbf{A}^T$. Determine the values of the constants α and β.
c. Show that the matrix $\mathbf{M}$ has the property $\mathbf{M}^T = -\mathbf{M}$. Matrices with this property are called skew-symmetric matrices.
d. If $\mathbf{v}$ is a right eigenvector of a skew-symmetric matrix $\mathbf{M}$ with eigenvalue μ, show that $\mathbf{v}^T$ is a left eigenvector with eigenvalue $-\mu$. Hence by considering the equality

$$\mu = \frac{\mathbf{v}^{*T} \mu \mathbf{v}}{\mathbf{v}^{*T} \mathbf{v}} = \frac{\mathbf{v}^{*T} \mathbf{M} \mathbf{v}}{\mathbf{v}^{*T} \mathbf{v}}$$

show that the complex conjugate of the eigenvalue is $\mu^* = -\mu$ and hence that all eigenvalues of a skew-symmetric matrix are imaginary.
e. Show that the dynamical system is stable if $\mathrm{Re}(\alpha + \beta\mu_r) < 0$ for all eigenvalues μ_r of the matrix $\mathbf{M}$, and hence that the condition for stability is simply $\alpha < 0$.

The last result means that if the individual vertices are stable in the absence of interaction with other vertices, then the coupled dynamical system is also stable at the symmetric fixed point.

18.4 Following the arguments of Section 18.2.2 the stability of a fixed point in certain dynamical systems on networks depends on the spectrum of eigenvalues of the adjacency matrix. Suppose we have a dynamical system on a network that takes the form of an $L \times L$ square lattice with periodic (toroidal) boundary conditions along its edges, and suppose we label each vertex of the lattice by its position vector $\mathbf{r} = (i, j)$ where $i, j = 1 \ldots L$ are the row and column indices of the vertex.

a. Consider the vector $\mathbf{v}$ with one element for each vertex such that $v_{\mathbf{r}} = \exp(\mathrm{i}\mathbf{k}^T\mathbf{r})$.

Show that this vector is an eigenvector of the adjacency matrix provided

$$\mathbf{k} = \frac{2\pi}{L}\begin{pmatrix} n_1 \\ n_2 \end{pmatrix},$$

where n_1 and n_2 are integers.

b. What range of values is permitted for the integers n_1 and n_2? Hence find the largest and smallest eigenvalues.

18.5 Consider a network with an oscillator on every vertex. The state of the oscillator on vertex i is represented by a phase angle θ_i and the system is governed by dynamical equations of the form

$$\frac{d\theta_i}{dt} = \omega + \sum_j A_{ij} g(\theta_i - \theta_j),$$

where ω is a constant and the function $g(x)$ respects the rotational symmetry of the phases, meaning that $g(x + 2\pi) = g(x)$ for all x.

a. Show that the synchronized state $\theta_i = \theta^* = \omega t$ for all i is a solution of the dynamics.

b. Consider a small perturbation away from the synchronized state $\theta_i = \theta^* + \in_i$ and show that the vector $\in = (\in_1, \in_2, ...)$ satisfies

$$\frac{d\epsilon}{dt} = g'(0)\,\mathbf{L}\epsilon,$$

where **L** is the graph Laplacian.

c. Hence show that the synchronized state is stable against small perturbations if and only if $g'(0) < 0$.